

112590024 邱紹澹 資工三

編譯方法

make

刪除

make clean

Ch 2

```
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024$ make all
make -C /lib/modules/6.19.8-200.fc43.x86_64/build M=/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024 modules
make[1]: 進入目錄「/usr/src/kernels/6.19.8-200.fc43.x86_64」
make[2]: 進入目錄「/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024」
  CC [M]  jiffies_proc.o
  MODPOST Module.symvers
  CC [M]  jiffies_proc.mod.o
  CC [M]  .module-common.o
  LD [M]  jiffies_proc.ko
  BTF [M] jiffies_proc.ko
Skipping BTF generation for jiffies_proc.ko due to unavailability of vmlinux
make[2]: 離開目錄「/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024」
make[1]: 離開目錄「/usr/src/kernels/6.19.8-200.fc43.x86_64」
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024$
```

2_1

```
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch2$ make
make -C /lib/modules/6.19.8-200.fc43.x86_64/build M=/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch2 modules
make[1]: 進入目錄「/usr/src/kernels/6.19.8-200.fc43.x86_64」
make[2]: 進入目錄「/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch2」
  CC [M]  jiffies_proc.o
  CC [M]  seconds_proc.o
  MODPOST Module.symvers
  LD [M]  jiffies_proc.ko
  BTF [M] jiffies_proc.ko
Skipping BTF generation for jiffies_proc.ko due to unavailability of vmlinux
  CC [M]  seconds_proc.mod.o
  LD [M]  seconds_proc.ko
  BTF [M] seconds_proc.ko
Skipping BTF generation for seconds_proc.ko due to unavailability of vmlinux
make[2]: 離開目錄「/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch2」
make[1]: 離開目錄「/usr/src/kernels/6.19.8-200.fc43.x86_64」
```

2_2_make

```
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch2$ sudo insmod jiffies_proc.ko
[sudo] kelen 的密碼:
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch2$ sudo insmod seconds_proc.ko
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch2$ cat /proc/jiffies
4295779241
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch2$
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch2$ cat /proc/seconds
18
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch2$ sleep 3
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch2$ cat /proc/jiffies
4295795883
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch2$ cat /proc/seconds
31
```

2_2_insmod

```
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch2$ sudo rmmod seconds_proc
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch2$ sudo rmmod jiffies_proc
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch2$ ls /proc/seconds
ls: 無法存取 '/proc/seconds': 沒有此一檔案或目錄
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch2$
```

2_2_rmmod

Ch 3.1

```
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3$ make clean
make -C /lib/modules/6.19.8-200.fc43.x86_64/build M=/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3 clean
make[1]: 進入目錄「/usr/src/kernels/6.19.8-200.fc43.x86_64」
make[2]: 進入目錄「/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3」
make[2]: 離開目錄「/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3」
make[1]: 離開目錄「/usr/src/kernels/6.19.8-200.fc43.x86_64」
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3$ make
make -C /lib/modules/6.19.8-200.fc43.x86_64/build M=/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3 modules
make[1]: 進入目錄「/usr/src/kernels/6.19.8-200.fc43.x86_64」
make[2]: 進入目錄「/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3」
CC [M] list_tasks.o
在從下述位置 include 的檔案: /usr/src/kernels/6.19.8-200.fc43.x86_64/include/asm-generic/bug.h:31,
          來自 /usr/src/kernels/6.19.8-200.fc43.x86_64/arch/x86/include/asm/bug.h:193,
          來自 /usr/src/kernels/6.19.8-200.fc43.x86_64/arch/x86/include/asm/alternative.h:9,
          來自 /usr/src/kernels/6.19.8-200.fc43.x86_64/arch/x86/include/asm/barrier.h:5,
          來自 /usr/src/kernels/6.19.8-200.fc43.x86_64/include/linux/list.h:11,
          來自 /usr/src/kernels/6.19.8-200.fc43.x86_64/include/linux/module.h:12,
          來自 list_tasks.c:2:
list_tasks.c: 在函式「list_tasks_init」中:
/usr/src/kernels/6.19.8-200.fc43.x86_64/include/linux/kern_levels.h:5:25: 警告: 格式「%ld」 expects argument of type「long int」, but argument 4 has type「unsigned int」 [-Wformat=]
   5 | #define KERN_SOH          "\001"          /* ASCII Start Of Header */
     |
```

Ch3_project1_1

```
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3/shell_build$ ./shell
hell
=====
=== Welcome to Simple UNIX Shell ===
=====
Commands: exit (quit), history (show history), cd (change directory)
Features: input/output redirection (<, >, >>), pipes (|)

osh> ls
Makefile  shell  shell.c
osh> pwd
/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3/shell_build
osh> date
西元2026年03月20日 (週五) 20時30分19秒 CST
osh> history

===== Command History =====
   1: ls
   2: pwd
   3: date
=====

osh> exit
Goodbye!
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3/shell_build$
```

Ch3_project1_2

```

kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3/shell_build$ cat .shell_history
ls
pwd
date
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3/shell_build$ ./shell
=====
=== Welcome to Simple UNIX Shell ===
=====
Commands: exit (quit), history (show history), cd (change directory)
Features: input/output redirection (<, >, >>), pipes (|)

osh> ls > output.txt
osh> cat output.txt
Makefile
output.txt
shell
shell.c
osh> echo "Line1" > test.txt
osh> echo "Line2" >> test.txt
osh> cat test.txt
"Line1"
"Line2"
osh> cat < test.txt
"Line1"
"Line2"

```

Ch3_project1_3

```

osh> ls | wc -l
6
osh> ls
file.txt  Makefile  output.txt  shell  shell.c  test.txt
osh>
osh> cd /tmp
osh> pwd
/tmp
osh> ls /nonexist
ls: 無法存取 '/nonexist': 沒有此一檔案或目錄

```

Ch3_project1_4

Ch3.2

```

kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3/Project2_TaskInfo$ make
make -C /lib/modules/6.19.8-200.fc43.x86_64/build M=/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3/Project2_TaskInfo modules
make[1]: 進入目錄「/usr/src/kernels/6.19.8-200.fc43.x86_64」
make[2]: 進入目錄「/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3/Project2_TaskInfo」
  CC [M] task_info.o
  MODPOST Module.symvers
  CC [M] task_info.mod.o
  CC [M] .module-common.o
  LD [M] task_info.ko
  BTF [M] task_info.ko
Skipping BTF generation for task_info.ko due to unavailability of vmlinux
make[2]: 離開目錄「/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3/Project2_TaskInfo」
make[1]: 離開目錄「/usr/src/kernels/6.19.8-200.fc43.x86_64」
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3/Project2_TaskInfo$ ls
Makefile      Module.symvers  task_info.ko   task_info.mod.c  task_info.o
modules.order  task_info.c     task_info.mod  task_info.mod.o

```

Ch3_project2_1

```

kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project2_TaskInfo$
sudo insmod task_info.ko
[sudo] kelen 的密碼:
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project2_TaskInfo$
cat /proc/task_info
=== Task Information ===
PID:      23066
PPID:     8844
Command:  cat
Priority:  120
State:    Running
=====
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project2_TaskInfo$
echo 1 > /proc/task_info
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project2_TaskInfo$
cat /proc/task_info
=== Task Information ===
PID:      1
PPID:     0
Command:  systemd
Priority:  120
State:    Interruptible
=====
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project2_TaskInfo$
sudo rmmod task_info
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project2_TaskInfo$
cat /proc/task_info
cat: /proc/task_info: 沒有此一檔案或目錄
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project2_TaskInfo$

```

Ch3_project2_2

Ch3.3

Part1

```

make[2]: 離開目錄「/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project3_ListingTasks」
make[1]: 離開目錄「/usr/src/kernels/6.19.8-200.fc43.x86_64」
make -C /lib/modules/6.19.8-200.fc43.x86_64/build M=/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project3_ListingTasks modules
make[1]: 進入目錄「/usr/src/kernels/6.19.8-200.fc43.x86_64」
make[2]: 進入目錄「/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project3_ListingTasks」
  CC [M]  list_tasks_linear.o
  CC [M]  list_tasks_dfs.o
MODPOST Module.symvers
  CC [M]  list_tasks_linear.mod.o
  CC [M]  .module-common.o
  LD [M]  list_tasks_linear.ko
  BTF [M] list_tasks_linear.ko
Skipping BTF generation for list_tasks_linear.ko due to unavailability of vmlinux
  CC [M]  list_tasks_dfs.mod.o
  LD [M]  list_tasks_dfs.ko
  BTF [M] list_tasks_dfs.ko
Skipping BTF generation for list_tasks_dfs.ko due to unavailability of vmlinux
make[2]: 離開目錄「/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project3_ListingTasks」
make[1]: 離開目錄「/usr/src/kernels/6.19.8-200.fc43.x86_64」
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project3_ListingTasks$ ls -la *.ko
-rw-r--r--. 1 kelen kelen 348888  3月 20 21:11 list_tasks_dfs.ko
-rw-r--r--. 1 kelen kelen 331752  3月 20 21:11 list_tasks_linear.ko
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project3_ListingTasks$ sudo insmod list_tasks_linear.ko
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project3_ListingTasks$ lsmod | grep list_tasks_linear
list_tasks_linear      12288  0

```

Ch3_project3_1

```
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3/Project3_ListingTasks$ sudo rmmod list_tasks_linear
lsmod | grep list_tasks_linear
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3/Project3_ListingTasks$ sudo dmesg | tail -10
[ 5032.013993] 27908 kworker/10:1 UNKNOWN
[ 5032.013994] 27909 psimon INTERRUPTIBLE
[ 5032.013994] 27914 dbus-update-act INTERRUPTIBLE
[ 5032.013995] 27915 sudo INTERRUPTIBLE
[ 5032.013995] 27916 insmod RUNNING
[ 5032.013996] 27917 9 RUNNING
[ 5032.013996] -----
[ 5032.013996] Total tasks found: 576
[ 5032.013997] =====
[ 5101.197128] Task listing module (linear) unloaded
```

Ch3_project3_2

Part2

```
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3/Project3_ListingTasks$ sudo insmod list_tasks_dfs.ko
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3/Project3_ListingTasks$ sleep 2
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3/Project3_ListingTasks$ sudo dmesg | head -80
[ 5238.569176] =====
[ 5238.569179] Task Lister - DFS Tree Traversal (Part II)
[ 5238.569180] =====
[ 5238.569180] Process Tree Structure:
[ 5238.569181] [ PID] Command State
[ 5238.569181] =====
[ 5238.569182] [ 0] swapper/0 RUNNING (root)
[ 5238.569183] └─ [ 1] systemd INTERRUPTIBLE
[ 5238.569184] │ └─ [ 675] systemd-journal INTERRUPTIBLE
[ 5238.569186] │ └─ [ 703] systemd-userdbd INTERRUPTIBLE
[ 5238.569187] │ └─ [28005] systemd-userwor INTERRUPTIBLE
[ 5238.569188] │ └─ [28006] systemd-userwor INTERRUPTIBLE
[ 5238.569190] │ └─ [28189] systemd-userwor INTERRUPTIBLE
[ 5238.569191] │ └─ [ 716] systemd-udev INTERRUPTIBLE
[ 5238.569192] │ └─ [ 969] systemd-oomd INTERRUPTIBLE
[ 5238.569193] │ └─ [ 970] systemd-resolve INTERRUPTIBLE
[ 5238.569194] │ └─ [ 971] auditd INTERRUPTIBLE
[ 5238.569194] │ └─ [ 975] sedispatch INTERRUPTIBLE
[ 5238.569195] │ └─ [1001] dbus-broker-lau INTERRUPTIBLE
[ 5238.569196] │ └─ [1003] dbus-broker INTERRUPTIBLE
[ 5238.569197] │ └─ [1006] avahi-daemon INTERRUPTIBLE
[ 5238.569198] │ └─ [1135] avahi-daemon UNKNOWN
[ 5238.569199] │ └─ [1010] irqbalance INTERRUPTIBLE
[ 5238.569200] │ └─ [1012] chronyd INTERRUPTIBLE
[ 5238.569201] │ └─ [1018] mcelog INTERRUPTIBLE
[ 5238.569202] │ └─ [1023] polkitd INTERRUPTIBLE
```

Ch3_project3_3

```
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_teambased_112590024/Ch3/Project3_ListingTasks$ sudo dmesg | tail -20
[ 5238.569667] └─ [27104] kworker/9:0 UNKNOWN
[ 5238.569668] └─ [27357] kworker/u81:2 UNKNOWN
[ 5238.569669] └─ [27410] kworker/19:1 UNKNOWN
[ 5238.569670] └─ [27475] kworker/18:0 UNKNOWN
[ 5238.569671] └─ [27503] kworker/u80:7 UNKNOWN
[ 5238.569672] └─ [27908] kworker/10:1 UNKNOWN
[ 5238.569672] └─ [27938] kworker/14:2 UNKNOWN
[ 5238.569673] └─ [28118] kworker/4:0 UNKNOWN
[ 5238.569674] └─ [28152] kworker/16:1 UNKNOWN
[ 5238.569675] └─ [28154] kworker/17:1 UNKNOWN
[ 5238.569676] └─ [28158] kworker/15:2 UNKNOWN
[ 5238.569677] └─ [28238] kworker/6:2 UNKNOWN
[ 5238.569677] └─ [28308] kworker/12:0 UNKNOWN
[ 5238.569678] └─ [28479] kworker/19:0 UNKNOWN
[ 5238.569679] └─ [28539] kworker/2:0 UNKNOWN
[ 5238.569680] └─ [28560] psimon INTERRUPTIBLE
[ 5238.569681] =====
```

Ch3_project3_4

```
[ 5372.656986] 29084      dbus-broker-lau      INTERRUPTIBLE
[ 5372.656986] 29091      dbus-broker          INTERRUPTIBLE
[ 5372.656987] 29092      sudo                INTERRUPTIBLE
[ 5372.656987] 29095      systemd-userwor      INTERRUPTIBLE
[ 5372.656988] 29096      kworker/4:3          UNKNOWN
[ 5372.656988] 29097      sudo                INTERRUPTIBLE
[ 5372.656989] 29098      insmod              RUNNING
[ 5372.656989] -----
[ 5372.656989] Total tasks found: 586
[ 5372.656990] =====
```

Ch3_project3_5

```
[ 5238.569676] | [28158] kworker/15:2      UNKNOWN
[ 5238.569677] | [28238] kworker/6:2        UNKNOWN
[ 5238.569677] | [28308] kworker/12:0       UNKNOWN
[ 5238.569678] | [28479] kworker/19:0       UNKNOWN
[ 5238.569679] | [28539] kworker/2:0        UNKNOWN
[ 5238.569680] | [28560] psimon           INTERRUPTIBLE
[ 5238.569681] =====
[ 5238.569681] Total tasks found: 560
[ 5238.569682] =====
[ 5238.569682] DFS tree traversal completed successfully!
[ 5372.656989] Total tasks found: 586
```

Ch3_project3_6

```
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project3_ListingTasks$
sudo rmmod list_tasks_dfs
sudo rmmod list_tasks_linear 2>/dev/null
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project3_ListingTasks$
lsmod | grep list_tasks
uname -r
6.19.8-200.fc43.x86_64
```

Ch3_project3_7

Ch3.4

Part1

```
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project4_KernelDataStructures$
make clean && make
make -C /lib/modules/6.19.8-200.fc43.x86_64/build M=/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project4_KernelDataStructures clean
make[1]: 進入目錄「/usr/src/kernels/6.19.8-200.fc43.x86_64」
make[2]: 進入目錄「/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project4_KernelDataStructures」
make[2]: 離開目錄「/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project4_KernelDataStructures」
make[1]: 離開目錄「/usr/src/kernels/6.19.8-200.fc43.x86_64」
make -C /lib/modules/6.19.8-200.fc43.x86_64/build M=/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project4_KernelDataStructures modules
make[1]: 進入目錄「/usr/src/kernels/6.19.8-200.fc43.x86_64」
make[2]: 進入目錄「/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project4_KernelDataStructures」
  CC [M]  linked_list.o
  CC [M]  collatz.o
  MODPOST Module.symvers
  CC [M]  linked_list.mod.o
  CC [M]  .module-common.o
  LD [M]  linked_list.ko
  BTF [M] linked_list.ko
Skipping BTF generation for linked_list.ko due to unavailability of vmlinux
  CC [M]  collatz.mod.o
  LD [M]  collatz.ko
  BTF [M] collatz.ko
Skipping BTF generation for collatz.ko due to unavailability of vmlinux
make[2]: 離開目錄「/home/kelen/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project4_KernelDataStructures」
make[1]: 離開目錄「/usr/src/kernels/6.19.8-200.fc43.x86_64」
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tebased_112590024/Ch3/Project4_KernelDataStructures$
sudo insmod linked_list.ko
```

Ch3_project4_1


```

kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tembased_112590024/Ch3/Project4_KernelDataStructures$
sudo dmesg | tail -30
[ 5551.080526] Task Lister DFS module unloaded.
[ 5551.139567] Task listing module (linear) unloaded
[ 6018.089666] === Linked List Module Loaded ===
[ 6018.089670]
    --- Traversing Linked List ---
[ 6018.089671] Color ID: 1, Name: Red, RGB: (255, 0, 0)
[ 6018.089673] Color ID: 2, Name: Green, RGB: (0, 255, 0)
[ 6018.089674] Color ID: 3, Name: Blue, RGB: (0, 0, 255)
[ 6018.089674] Color ID: 4, Name: Yellow, RGB: (255, 255, 0)
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tembased_112590024/Ch3/Project4_KernelDataStructures$
sudo rmmod linked_list
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tembased_112590024/Ch3/Project4_KernelDataStructures$
sudo dmesg | tail -15
[ 6018.089673] Color ID: 2, Name: Green, RGB: (0, 255, 0)
[ 6018.089674] Color ID: 3, Name: Blue, RGB: (0, 0, 255)
[ 6018.089674] Color ID: 4, Name: Yellow, RGB: (255, 255, 0)
[ 6018.089675] --- End of List Traversal ---

[ 6104.744133]
    === Linked List Module Unloading ===
[ 6104.744136] --- Removing all elements from the list ---
[ 6104.744137] Removing: ID 1, Name: Red
[ 6104.744138] Removing: ID 2, Name: Green
[ 6104.744139] Removing: ID 3, Name: Blue
[ 6104.744140] Removing: ID 4, Name: Yellow
[ 6104.744140] --- All elements removed, memory freed ---
[ 6104.744141] === Linked List Module Unloaded ===

```

Ch3_project4_2

Part2

```

kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tembased_112590024/Ch3/Project4_KernelDataStructures$
sudo insmod collatz.ko
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tembased_112590024/Ch3/Project4_KernelDataStructures$
sudo dmesg | tail -40
[ 6253.446814] === Collatz Sequence Module Loaded ===
[ 6253.446817] === Generating Collatz Sequence for n = 10 ===
[ 6253.446819]
    --- Traversing Collatz Sequence ---
[ 6253.446819] Step 1: 10
[ 6253.446820] Step 2: 5
[ 6253.446821] Step 3: 16
[ 6253.446822] Step 4: 8
[ 6253.446822] Step 5: 4
[ 6253.446823] Step 6: 2
[ 6253.446823] Step 7: 1
[ 6253.446824] --- Total steps in sequence: 7 ---

```

Ch3_project4_3

```

kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tembased_112590024/Ch3/Project4_KernelDataStructures$
sudo rmmod collatz
sudo dmesg -c
sudo insmod collatz.ko initial_value=7
sudo dmesg | tail -40

```

Ch3_project4_4

```

[ 6285.384593] === Collatz Sequence Module Loaded ===
[ 6285.384596] === Generating Collatz Sequence for n = 7 ===
[ 6285.384599]
[ 6285.384599] --- Traversing Collatz Sequence ---
[ 6285.384599] Step 1: 7
[ 6285.384600] Step 2: 22
[ 6285.384601] Step 3: 11
[ 6285.384601] Step 4: 34
[ 6285.384602] Step 5: 17
[ 6285.384602] Step 6: 52
[ 6285.384603] Step 7: 26
[ 6285.384603] Step 8: 13
[ 6285.384604] Step 9: 40
[ 6285.384604] Step 10: 20
[ 6285.384605] Step 11: 10
[ 6285.384605] Step 12: 5
[ 6285.384606] Step 13: 16
[ 6285.384606] Step 14: 8
[ 6285.384607] Step 15: 4
[ 6285.384607] Step 16: 2
[ 6285.384608] Step 17: 1
[ 6285.384608] --- Total steps in sequence: 17 ---

```

Ch3_project4_5

```

kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tembased_112590024/Ch3/Project4_KernelDataStructures$
sudo rmmod collatz
kelen@host:~/文件/GitHub/OSHW/HW1/OSHW#1_tembased_112590024/Ch3/Project4_KernelDataStructures$
sudo dmesg | tail -15
[ 6285.384605] Step 11: 10
[ 6285.384605] Step 12: 5
[ 6285.384606] Step 13: 16
[ 6285.384606] Step 14: 8
[ 6285.384607] Step 15: 4
[ 6285.384607] Step 16: 2
[ 6285.384608] Step 17: 1
[ 6285.384608] --- Total steps in sequence: 17 ---

[ 6348.447762]
[ 6348.447762] === Collatz Sequence Module Unloading ===
[ 6348.447765] --- Removing all sequence elements ---
[ 6348.447767] --- All elements removed, memory freed ---
[ 6348.447768] === Collatz Sequence Module Unloaded ===

```

Ch3_project4_6